Rev E: rear USB hub, PC data, USB-C power and watch charging

Rev E. USB and power circuitry. Read CHECKS.md for validation status.

USB2517 self-powered USB 2.0 hub

File: hub.kicad_sch

Hub clock and supply

File: hub_supply.kicad_sch

PC upstream USB-C

File: pc_usb.kicad_sch

USB port enable interlock

File: port_enables.kicad_sch

External USB-A 1

File: usb_a_1.kicad_sch

External USB-A 2

File: usb_a_2.kicad_sch

External USB-A 3

File: usb_a_3.kicad_sch

USB-C watch output

File: watch.kicad_sch

USB-C charging input and PD negotiation

File: pd_input.kicad_sch

Power and control harness

File: harness.kicad_sch

Internal CM4 USB device port power

File: cm4_usb_power.kicad_sch

USB2517 self-powered USB 2.0 hub — continued 2

File: hub_2.kicad_sch

Independent USB hub supply

File: usb_service_power.kicad_sch

Combined charge/data and alternate host selection

File: usb_upstream_select.kicad_sch

Safe upstream attach detection

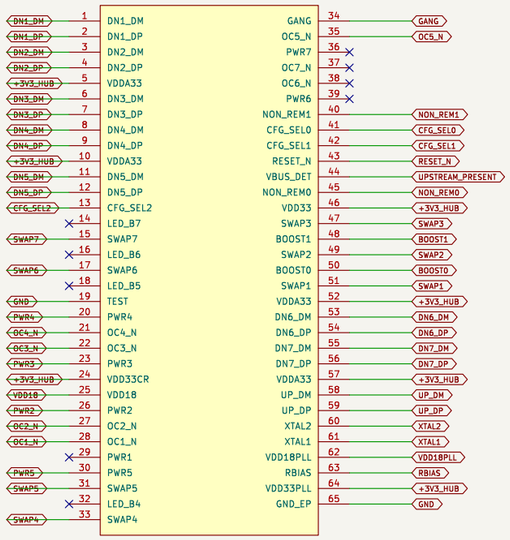
File: usb_attach_sense.kicad_sch

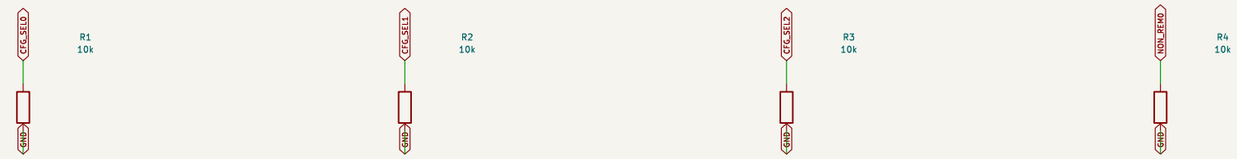
Sheet: /
File: usb-io.kicad_sch

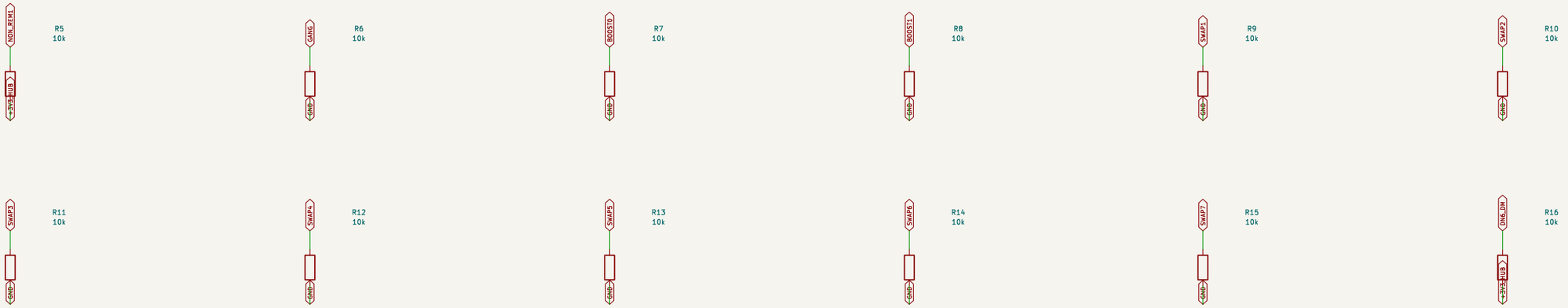
Title:

Size: A2	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 16/16

Five enabled ports: 1 keyboard, 2 CM4 USB device, 3–5 external USB-A.
CFG=000; individual power switching; ports 1/2 non-removable; 6/7 disabled. No upstream VBUS backfeed.







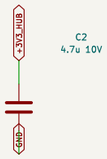
Connected schematic; hardware and manufacturing validation pending

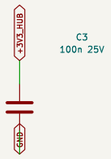
Sheet: /USB2517 self-powered USB 2.0 hub/
File: hub.kicad_sch

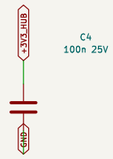
Title: USB2517 self-powered USB 2.0 hub

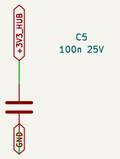
Size: A3	Date:	**Rev: G**
KiCad E.D.A. 9.0.7		Id: 2/16

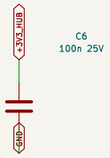
24 MHz crystal, internal 1.8 V regulators, 3.3 V LDO and local bypass.
No series termination on USB2517 data pins; integrated termination is used.

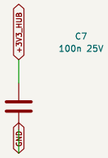


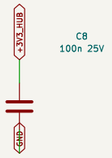


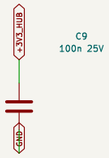




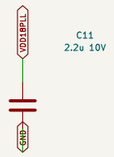




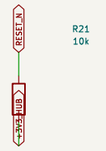


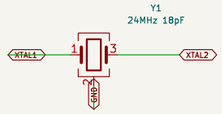
















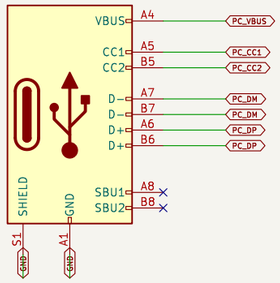
Connected schematic; hardware and manufacturing validation pending

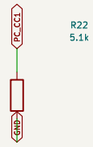
Sheet: /Hub clock and supply/
File: hub_supply.kicad_sch

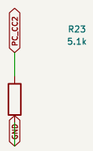
Title: Hub clock and supply

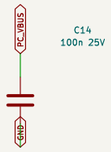
Size: A3	Date:	**Rev: G**
KiCad E.D.A. 9.0.7		Id: 3/16

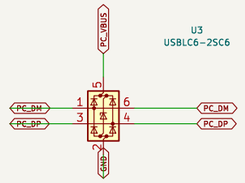
USB 2.0 UFP. CC1/CC2 each have 5.1 kOhm Rd. PC_VBUS is sensing only.
Shells connect to chassis/ground; connector data pairs require a continuous ground reference.





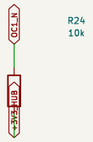


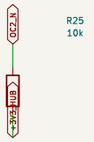












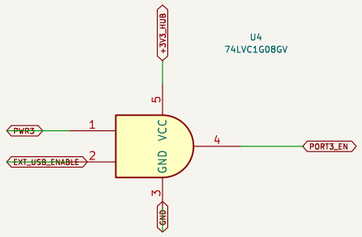
Connected schematic; hardware and manufacturing validation pending

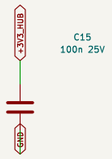
Sheet: /PC upstream USB-C/
File: pc_usb.kicad_sch

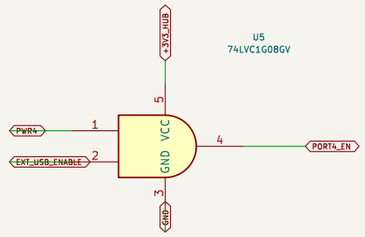
Title: PC upstream USB-C

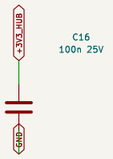
Size: A3 | Date: | **Rev: G**
KiCad E.D.A. 9.0.7 | Id: 4/16

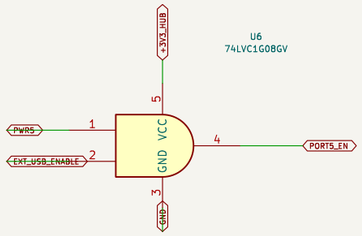
The hub controls per-port VBUS. EXT_USB_ENABLE must also be high.
A low or unprogrammed enable keeps all three external USB-A supplies off.

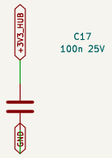












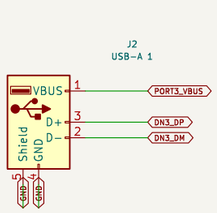
Connected schematic; hardware and manufacturing validation pending

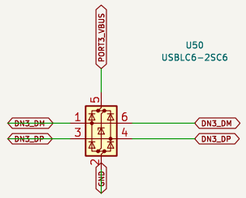
Sheet: /USB port enable interlock/
File: port_enables.kicad_sch

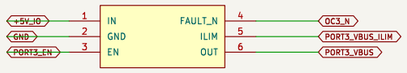
Title: USB port enable interlock

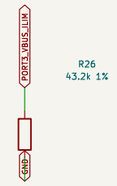
Size: A3	Date:	**Rev: G**
KiCad E.D.A. 9.0.7		Id: 5/16

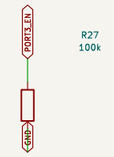
USB 2.0 downstream port 3: 500 mA permitted load. TPS2553 overcurrent reports to hub.
150 uF minimum port reservoir; fault limits include resistor/device tolerance.

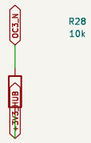


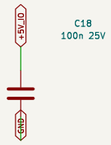


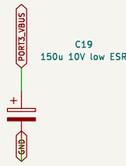


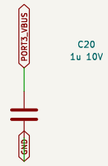












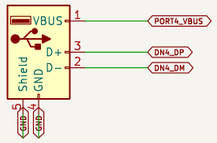
Connected schematic; hardware and manufacturing validation pending

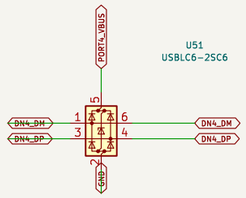
Sheet: /External USB-A 1/
File: usb_a_1.kicad_sch

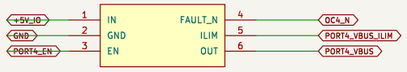
Title: External USB-A 1

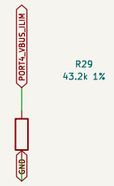
Size: A3	Date:	**Rev: G**
KiCad E.D.A. 9.0.7		Id: 6/16

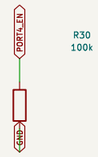
USB 2.0 downstream port 4: 500 mA permitted load. TPS2553 overcurrent reports to hub.
150 uF minimum port reservoir; fault limits include resistor/device tolerance.

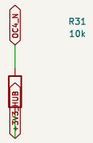


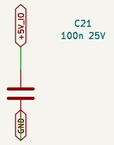


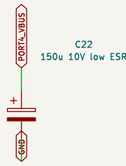


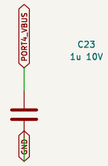












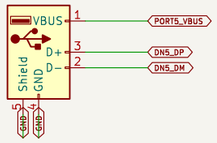
Connected schematic; hardware and manufacturing validation pending

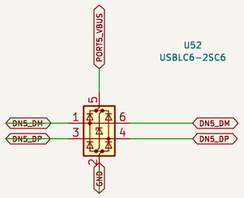
Sheet: /External USB-A 2/
File: usb_a_2.kicad_sch

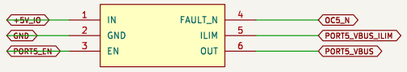
Title: External USB-A 2

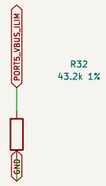
Size: A3	Date:	**Rev: G**
KiCad E.D.A. 9.0.7		Id: 7/16

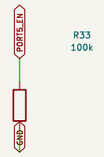
USB 2.0 downstream port 5: 500 mA permitted load. TPS2553 overcurrent reports to hub.
150 uF minimum port reservoir; fault limits include resistor/device tolerance.

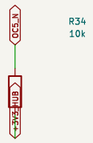


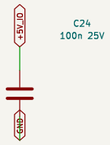


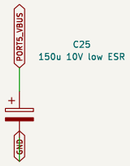


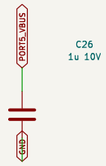












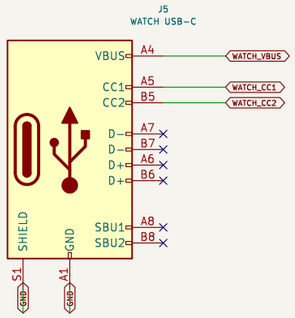
Connected schematic; hardware and manufacturing validation pending

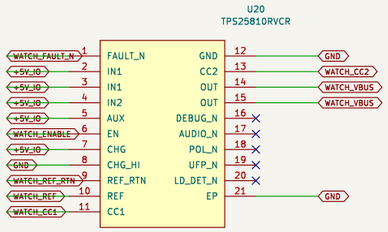
Sheet: /External USB-A 3/
File: usb_a_3.kicad_sch

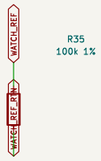
Title: External USB-A 3

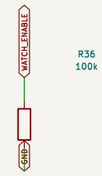
Size: A3	Date:	**Rev: G**
KiCad E.D.A. 9.0.7		Id: 8/16

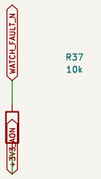
TPS25810: 5 V, 1.5 A advertised. CC attach detection, current limit, VCONN, discharge.
Charge-only output for existing Samsung USB-C watch puck. Fast-charge behavior needs a hardware test.

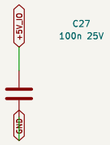


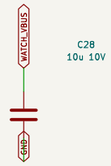


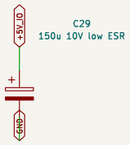












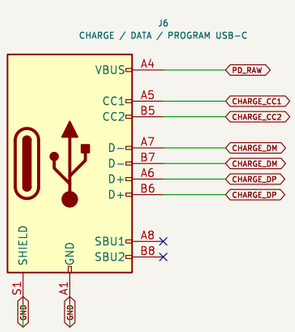
Connected schematic; hardware and manufacturing validation pending

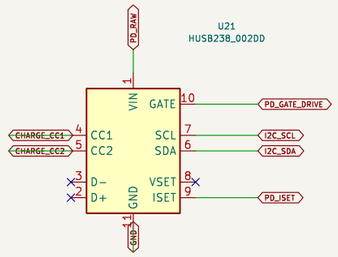
Sheet: /USB-C watch output/
File: watch.kicad_sch

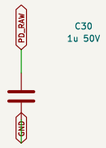
Title: USB-C watch output

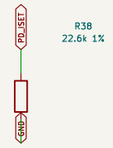
Size: A3	Date:	**Rev: G**
KiCad E.D.A. 9.0.7		Id: 9/16

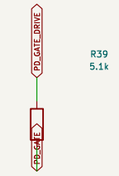
HUSB238 requests up to 20 V / 3 A. BQ25798 input current stays conservative until firmware validates contract.
Back-to-back PMOS isolate the input in both directions. 12 V gate clamp protects VGS.

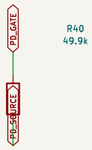


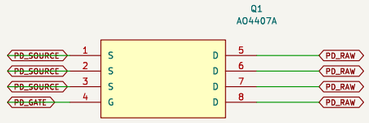


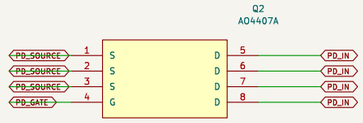




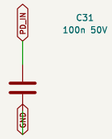


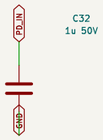












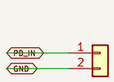
Connected schematic; hardware and manufacturing validation pending

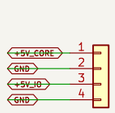
Sheet: /USB−C charging input and PD negotiation/
File: pd_input.kicad_sch

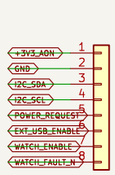
Title: USB−C charging input and PD negotiation

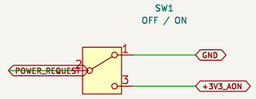
Size: A3	Date:	**Rev: G**
KiCad E.D.A. 9.0.7		Id: 10/16

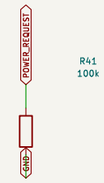
Rear switch is a logic request. Firmware completes Linux shutdown before releasing MAIN_ENABLE.
J9/J10 use separate keyed power harnesses; signals use a separate GH harness. No raw battery here.

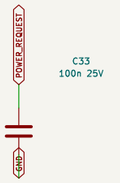




































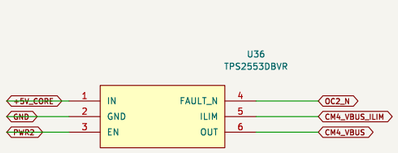
Connected schematic; hardware and manufacturing validation pending

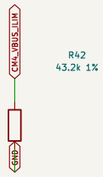
Sheet: /Power and control harness/
File: harness.kicad_sch

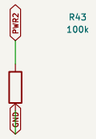
Title: Power and control harness

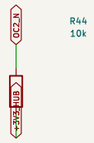
Size: A3	Date:	**Rev: G**
KiCad E.D.A. 9.0.7		Id: 11/16

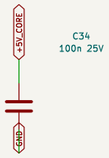
Switched 5 V VBUS follows hub port 2. CM4 is self-powered from its core rail.
The carrier must sense this VBUS through a suitable divider; never connect 5 V directly to GPIO.

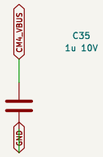


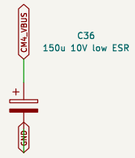












Connected schematic; hardware and manufacturing validation pending

Sheet: /Internal CM4 USB device port power/
File: cm4_usb_power.kicad_sch

Title: Internal CM4 USB device port power

Size: A3	Date:	**Rev: G**
KiCad E.D.A. 9.0.7		Id: 12/16

Five enabled ports: 1 keyboard, 2 CM4 USB device, 3–5 external USB-A.
CFG=000; individual power switching; ports 1/2 non-removable; 6/7 disabled. No upstream VBUS backfeed.



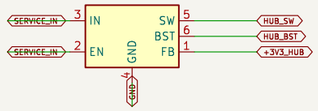
Connected schematic; hardware and manufacturing validation pending

Sheet: /USB2517 self-powered USB 2.0 hub – continued 2/
File: hub_2.kicad_sch

Title: USB2517 self-powered USB 2.0 hub – continued 2

Size: A3	Date:	**Rev: G**
KiCad E.D.A. 9.0.7		Id: 13/16

AP63203 fixed 3.3 V. Sources are isolated by 40 V Schottky diodes.
The hub stays powered from the connected host/charger during RP2040 reset. Do not connect sources directly.

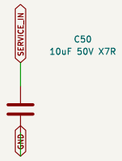


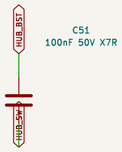


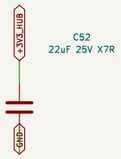


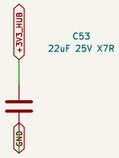
















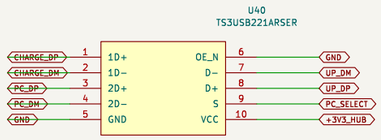
Connected schematic; hardware and manufacturing validation pending

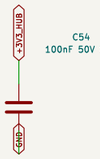
Sheet: /Independent USB hub supply/
File: usb_service_power.kicad_sch

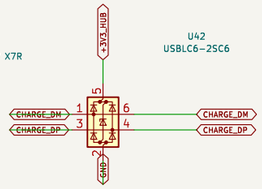
Title: Independent USB hub supply

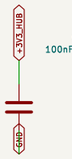
Size: A3	Date:	**Rev: F**
KiCad E.D.A. 9.0.7		Id: 14/16

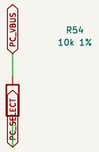
Port 1: charging USB-C data. Port 2: existing PC DATA connector. PC takes priority when both are plugged in.
Only data is selected; VBUS rails stay isolated. HUSB238 DP/DM are deliberately not connected; USB PD remains on CC1/CC2.

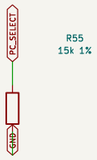












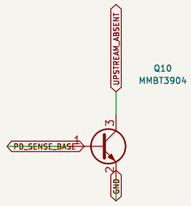
Connected schematic; hardware and manufacturing validation pending

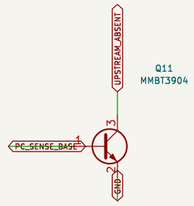
Sheet: /Combined charge/data and alternate host selection/
File: usb_upstream_select.kicad_sch

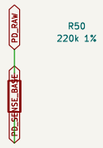
Title: Combined charge/data and alternate host selection

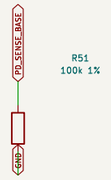
Size: A3	Date:	**Rev: F**
KiCad E.D.A. 9.0.7		Id: 15/16

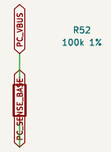
Two NPN collectors implement OR presence detection without coupling source VBUS rails.
3.3 V logic drives hub VBUS_DET; the PD input can be 20 V and must never connect directly to this pin.

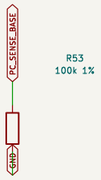


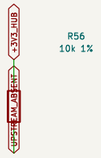


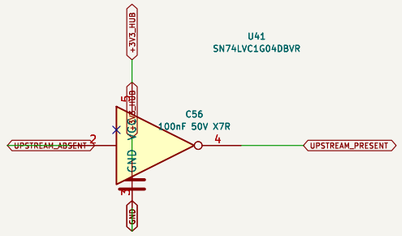












Connected schematic; hardware and manufacturing validation pending

Sheet: /Safe upstream attach detection/
File: usb_attach_sense.kicad_sch

Title: Safe upstream attach detection

Size: A3	Date:	**Rev: F**
KiCad E.D.A. 9.0.7		Id: 16/16